ASSIGNMENT No. 01
Basic Input/Output, Algorithm, Pseudocode & Flowchart: *Develop algorithms, pseudocode, and flowcharts for real-world problems (e.g., Simple Interest, Area of Circle, Temperature Conversion), and implement them in C using formatted input/output statements.*

1. Write a program (WAP) to display "Hello World" on console display.

```
// Header file for input output functions
#include <stdio.h>
// Main function: entry point for execution
int main()
{
// Writing print statement to print hello world
printf("Hello World");
return 0;
}
```

2. WAP to input an integer and display it.

```
// C Program to Print
#include <stdio.h>
// Driver code
int main()
{
// Declaring integer
int x = 5;
// Printing values
printf("Printing Integer value %d", x);
return 0;
}
```

3. WAP to input a float number and display it.
4. WAP to input a character and display it.
5. WAP to add, subtract, multiply, divide numbers.

```
// C program to add two numbers
#include <stdio.h>
int main() {
int a, b, sum = 0;
// Read two numbers from the user
printf("Enter two integers: ");
scanf("%d %d", &a, &b);
// Calculate the addition of a and b
// using '+' operator
sum = a + b;
printf("Sum: %d", sum);
return 0;
}
```

6. WAP to input a fraction (rational number) and display. (Ask the user to input numerator and denominator, then display it in the form of p/q without simplification).
7. WAP to Print the ASCII Value of a Character.

Using Format Specifier – Implicit

```
// C program to print ASCII Value of Character using
// implicit conversion with format specifier.
#include <stdio.h>
int main() {
char c = 'k';
// %d displays the integer value of
// a character
// %c displays the actual character
printf("The ASCII value of %c is %d", c, c);
return 0;
}
```

Using Typecasting - Explicit

```
// C program to print ASCII Value of Character using
// typecasting into integer
#include <stdio.h>
int main() {
char ch = 'A';
// Find the ASCII value of a character using typecasting
int asciiValue = (int)ch;
printf("ASCII value of %c is %d\n", ch, asciiValue);
return 0;
}
```

8. WAP to Swap Two Numbers.

```
#include <stdio.h>
int main() {
int a = 5, b = 10, temp;
// Swapping values of a and b
temp = a;
a = b;
b = temp;
printf("a = %d, b = %d\n", a, b);
return 0;
}
```

9. WAP for the area of a circle.
10. WAP for simple interest.
11. WAP for compound interest.
12. WAP to calculate gross salary of a person, where gross_salary=basic+da+ta and da is 10% of basic and ta is 12% of basic.

13. Write a C Program to Find the Size of int, float, double and char.

```
// C Program to Find the Size of int, float, double, and
// char using sizeof operator directly
#include <stdio.h>
int main() {
// Determine and Print the size of int
printf("Size of int: %u bytes\n", sizeof(int));
// Determine and Print the size of float
printf("Size of float: %u bytes\n", sizeof(float));
// Determine and Print the size of double
printf("Size of double: %u bytes\n", sizeof(double));
// Determine and Print the size of char
printf("Size of char: %u bytes\n", sizeof(char));
return 0;
}
```

14. WAP to Convert Fahrenheit to Celcius in C.

```
// Fahrenheit to Celsius
#include <stdio.h>
// Function to convert Degree
// Fahrenheit to Degree Celsius
float fahrenheit_to_celsius(float f)
{
return ((f - 32.0) * 5.0 / 9.0);
}
// Driver code
int main()
{
float f = 40;
// Passing parameter to function
printf("Temperature in Degree Celsius : %0.2f",
fahrenheit_to_celsius(f));
return 0;
}
```

15. You plan a trip to Europe costing ₹1,50,000. You take a travel loan from a bank at a simple interest rate of 8% per annum for 2 years. Write a C program to calculate the total interest you will pay and the total amount to be repaid at the end of the term.

16. You have a circular garden in your backyard with a radius of 7.5 meters. You want to cover the garden with grass turf, and the turf is sold at ₹55 per square meter. Write a C program to calculate the area of the garden and the total cost of covering it with turf.

17. Assume you are at the top of Redfort in Delhi. The height of Redfort is 17 m and you are looking towards point C which is located at road in diagonal direction. From you to point C the distance is 25 cm (in diagonal direction). Write a C program that calculates the distance between base of Redfort to point C.